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My New Way of Working with AI Tooling

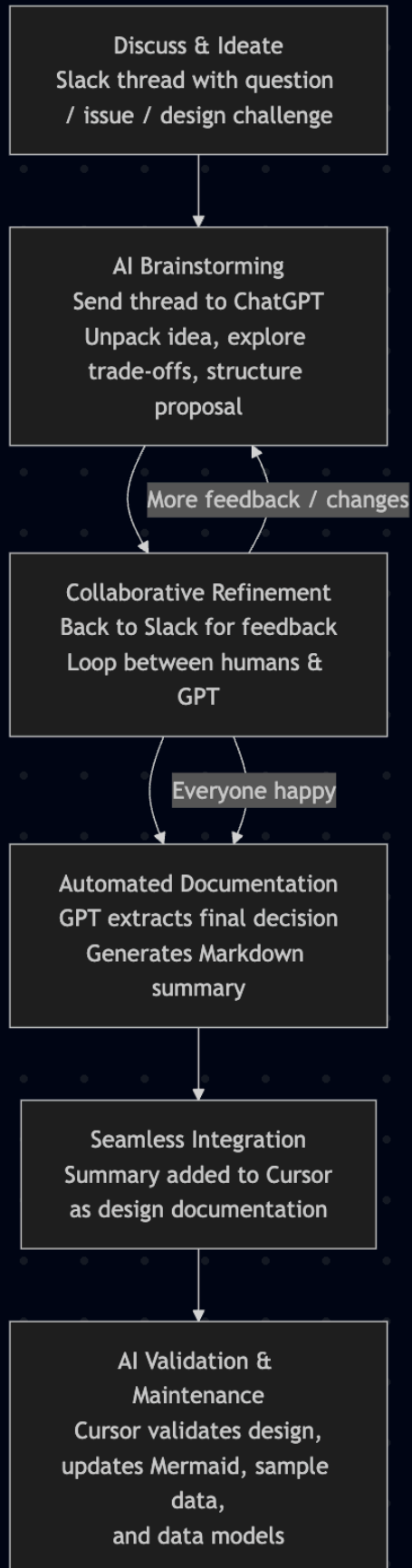
Since adopting **ChatGPT**, **Cursor**, and **GitHub Spec Kit**, our development workflow has changed beyond recognition.

The combination of these tools has created a seamless, high-speed loop between ideation, design, validation, and documentation. It feels like the future of engineering, and honestly, it's hard to imagine going back.

How It Works

1. **Discuss & Ideate** – Everything starts on Slack. Someone raises a question, an issue, or a design challenge.
2. **AI Brainstorming** – I take the thread to **ChatGPT** to unpack the idea, explore trade-offs, and shape it into a structured proposal.
3. **Collaborative Refinement** – The refined idea goes back into Slack for feedback. We loop between human input and GPT iterations until everyone's happy.
4. **Automated Documentation** – GPT then extracts the final decision and generates a **Markdown summary**.
5. **Seamless Integration** – The summary gets added directly into our **Cursor design documentation**.
6. **AI Validation & Maintenance** – Cursor cross-checks the new decision against the rest of the design, updates **Mermaid diagrams**, sample data, and data models automatically.

The whole loop—discussion to documented decision—takes **5–10 minutes**.



All Our Design Work, Centralized

Everything now lives in Cursor:

- Meeting notes
- Design discussions
- Data models
- Business rules
- Sample data
- Mermaid process and data flow diagrams

Cursor doesn't just store—it *thinks*. It constantly restructures my documents, surfaces related concepts, and identifies connections I wouldn't have noticed. It's often *miles ahead of me*.

During design reviews, it's become a running joke: I'll suggest adding something, and someone will reply, "No, James—it's already there." And sure enough, it is—perfectly phrased and in the right place.

For the design-heavy phase I'm in, this setup is **saving weeks, even months**, by maintaining a living, self-updating documentation set. It flags inconsistencies, highlights conflicting design decisions, and even reminds us about unfinished actions.

I also wrote a **script that merges all Markdown files into a single .md** so I can upload the entire corpus to my **custom GPT** for deeper analysis and synthesis.

AI-Assisted Meeting Notes Review

Even our meeting notes go through AI review. I have GPT check every note for tone, professionalism, and objectivity.

All Clear

All notes are factual, concise, and appropriate. They include:

- Technical summaries

- Task assignments
- Design and architecture decisions
- Professional references to team members

 No Issues Found with:

- Inappropriate or biased language
- Personal remarks
- Controversial opinions

The notes are stakeholder-ready, maintaining a **neutral, business-like tone** that reflects professionalism and focus.

The Real Breakthrough: Cognitive Flow

The real transformation is in **mental flow**. There's no context switching. The iteration loop is so fast that I can make complex design decisions **in real time** while the problem is still fresh in my head.

Before, I'd constantly postpone things—"I'll come back to that later." Now, those delays are gone. It's like working in a state of continuous clarity.

AI Racing Ahead

For the first time, I realised how quickly AI is pulling ahead of us.

It's suggesting **best practices and design improvements** that are absolutely correct—but we often can't implement them immediately because we're still finishing previously committed work.

It's a strange feeling: **I'm the bottleneck now**. The AI is moving faster than my human brain can deliver.

This has made me think seriously about **automating the next stage**:

- Developing a **code refactoring automation** system

- Building a **regression suite** the AI can call autonomously
- Letting it **run overnight tasks** (e.g., refactoring designs or verifying logic) so I wake up to a fresh, refactored system

The challenge right now is **verbosity**—AI produces so much documentation that I spend time compacting, deduplicating, and structuring it across multiple docs. I'm experimenting with ways to manage this better.

Managing Multiple Design States

I'm currently maintaining **three versions of the design**:

1. **Current** – the implemented, production-aligned version
2. **Next / Immediate Future** – improvements currently in flight
3. **Long-Term / Ideal End State** – the ultimate architecture vision

We're considering using **feature branches** or **subfolder structures** to manage these versions cleanly. Each has its own lifecycle, dependencies, and assumptions—but the AI is already starting to help reconcile differences between them.

The Result

This AI-augmented workflow has reshaped not just how we work, but how we *think*:

- **Instant iteration loops** with almost zero mental drag
- **Automated documentation, validation, and visualization**
- **Proactive surfacing of design insights and inconsistencies**
- **AI-driven suggestions that extend beyond human bandwidth**

It's efficient, creative, and constantly evolving—and even though it's already miles ahead of me, I'm doing my best to keep up.
